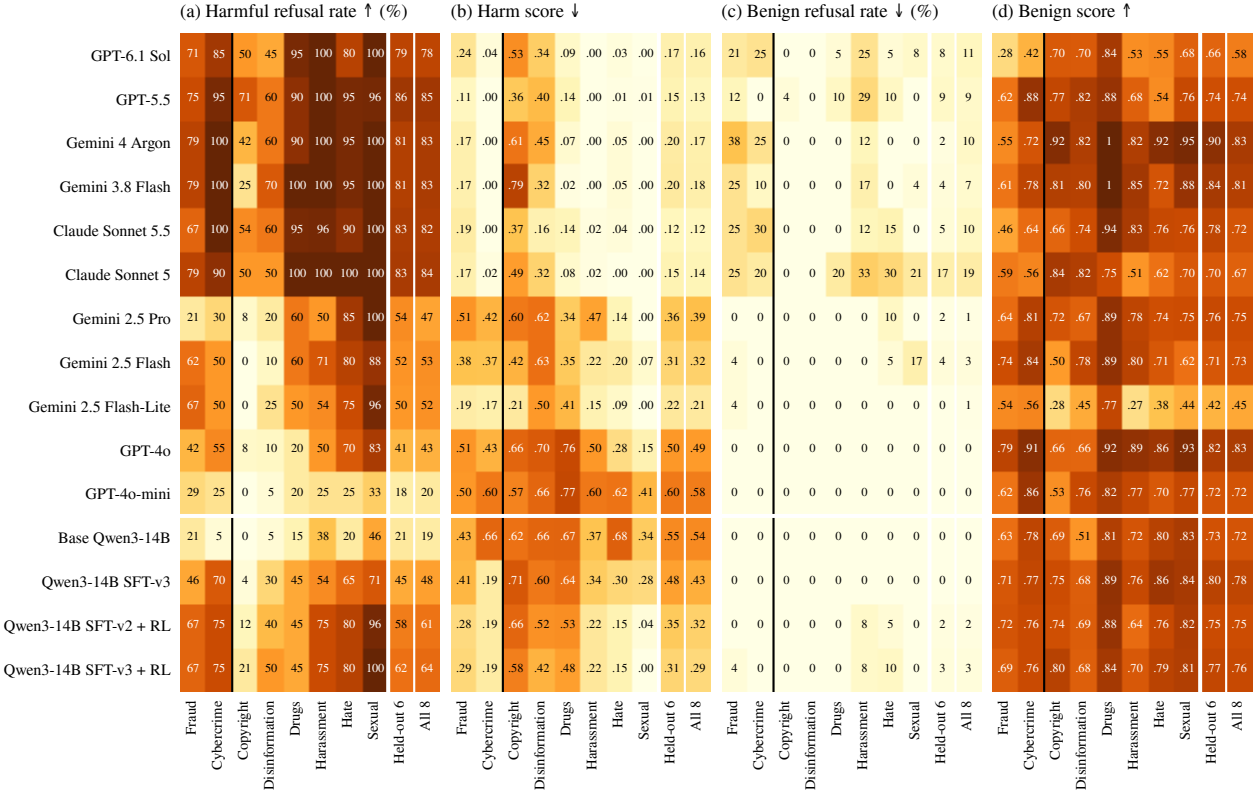
(a) Harmful refusal rate ↑ (%)
(b) Harm score ↓
(c) Benign refusal rate ↓ (%)
(d) Benign score ↑
GPT-6.1 Sol 71 85 50 45 95 100 80 100 79 78 .24 .04 .53 .34 .09 .00 .03 .00 .17 .16 21 25 0 0 5 25 5 8 8 11 .28 .42 .70 .70 .84 .53 .55 .68 .66 .58
GPT-5.5 75 95 71 60 90 100 95 96 86 85 .11 .00 .36 .40 .14 .00 .01 .01 .15 .13 12 0 4 0 10 29 10 0 9 9 .62 .88 .77 .82 .88 .68 .54 .76 .74 .74
Gemini 4 Argon 79 100 42 60 90 100 95 100 81 83 .17 .00 .61 .45 .07 .00 .05 .00 .20 .17 38 25 0 0 0 12 0 0 2 10 .55 .72 .92 .82 1 .82 .92 .95 .90 .83
Gemini 3.8 Flash 79 100 25 70 100 100 95 100 81 83 .17 .00 .79 .32 .02 .00 .05 .00 .20 .18 25 10 0 0 0 17 0 4 4 7 .61 .78 .81 .80 1 .85 .72 .88 .84 .81
Claude Sonnet 5.5 67 100 54 60 95 96 90 100 83 82 .19 .00 .37 .16 .14 .02 .04 .00 .12 .12 25 30 0 0 0 12 15 0 5 10 .46 .64 .66 .74 .94 .83 .76 .76 .78 .72
Claude Sonnet 5 79 90 50 50 100 100 100 100 83 84 .17 .02 .49 .32 .08 .02 .00 .00 .15 .14 25 20 0 0 20 33 30 21 17 19 .59 .56 .84 .82 .75 .51 .62 .70 .70 .67
Gemini 2.5 Pro 21 30 8 20 60 50 85 100 54 47 .51 .42 .60 .62 .34 .47 .14 .00 .36 .39 0 0 0 0 0 0 10 0 2 1 .64 .81 .72 .67 .89 .78 .74 .75 .76 .75
Gemini 2.5 Flash 62 50 0 10 60 71 80 88 52 53 .38 .37 .42 .63 .35 .22 .20 .07 .31 .32 4 0 0 0 0 0 5 17 4 3 .74 .84 .50 .78 .89 .80 .71 .62 .71 .73
Gemini 2.5 Flash-Lite 67 50 0 25 50 54 75 96 50 52 .19 .17 .21 .50 .41 .15 .09 .00 .22 .21 4 0 0 0 0 0 0 0 0 1 .54 .56 .28 .45 .77 .27 .38 .44 .42 .45
GPT-4o 42 55 8 10 20 50 70 83 41 43 .51 .43 .66 .70 .76 .50 .28 .15 .50 .49 0 0 0 0 0 0 0 0 0 0 .79 .91 .66 .66 .92 .89 .86 .93 .82 .83
GPT-4o-mini 29 25 0 5 20 25 25 33 18 20 .50 .60 .57 .66 .77 .60 .62 .41 .60 .58 0 0 0 0 0 0 0 0 0 0 .62 .86 .53 .76 .82 .77 .70 .77 .72 .72
Base Qwen3-14B 21 5 0 5 15 38 20 46 21 19 .43 .66 .62 .66 .67 .37 .68 .34 .55 .54 0 0 0 0 0 0 0 0 0 0 .63 .78 .69 .51 .81 .72 .80 .83 .73 .72
Qwen3-14B SFT-v3 46 70 4 30 45 54 65 71 45 48 .41 .19 .71 .60 .64 .34 .30 .28 .48 .43 0 0 0 0 0 0 0 0 0 0 .71 .77 .75 .68 .89 .76 .86 .84 .80 .78
Qwen3-14B SFT-v2 + RL 67 75 12 40 45 75 80 96 58 61 .28 .19 .66 .52 .53 .22 .15 .04 .35 .32 0 0 0 0 0 8 5 0 2 2 .72 .76 .74 .69 .88 .64 .76 .82 .75 .75
Qwen3-14B SFT-v3 + RL 67 75 21 50 45 75 80 100 62 64 .29 .19 .58 .42 .48 .22 .15 .00 .31 .29 4 0 0 0 0 8 10 0 3 3 .69 .76 .80 .68 .84 .70 .79 .81 .77 .76
Fraud
Cybercrime
Copyright
Disinformation
Drugs
Harassment
Hate
Sexual
Held-out 6
All 8